Decent Coaching Classes
VIII SB
Academic Conquest - VII

Subject: General Science (Ch. 5 to 6) **Answer Key | Marks: 30**

Q.1 Choose the correct alternative. (1 mark each) **[5]**

1. (b) 6 [$n = A - Z = 12 - 6$]
2. (b) 8 [$2n^2 = 2 \times 2^2$]
3. (a) 1 [one electron short of octet]
4. (b) Salt solution
5. (a) solid

Q.2 Fill in the blanks. (1 mark each) **[4]**

(a) atomic
(b) isotopes
(c) valence
(d) compound

Q.3 Do as directed. **[6]**

(A) True / False (1 mark each)

(a) True.
(b) False. A neutron is electrically neutral (a proton is positively charged).
(c) False. The proportion of constituents in a mixture can change; it is constant in a compound.

(B) Match the pairs (1 mark each)

1 → c (J. J. Thomson) 2 → a (Rutherford) 3 → b (Niels Bohr)

Q.4 Scientific reasons (any 2) (2 marks each) **[4]**

(a) The valence shell of helium is completely filled (electron duplet) and that of neon has eight electrons (electron octet). Their valency is zero, so they do not form chemical bonds.

(b) Water is a compound formed by chemical combination of hydrogen and oxygen. The properties of a compound are different from those of its constituent elements. Water is a liquid that is neither inflammable nor does it support combustion, so it extinguishes fire.

(c) In a solid the intermolecular force of attraction is strong and the particles are very close to each other, vibrating about fixed positions. Hence a solid has a definite shape and volume.

Q.5 Short answers (any 3) (2 marks each) **[6]**

(a) Protons = atomic number = 12; electrons = 12; neutrons = $A - Z = 24 - 12 = 12$ (1 mark). Electronic configuration: 2, 8, 2 (1 mark).

(b) Any two (1 mark each): Uranium-235 is used for nuclear fission and electricity generation; Cobalt-60 is used in treatment of cancer; Iodine-131 is used in treatment of goiter; Sodium-24 is used to detect leaks in underground pipes; isotopes preserve food from microbes; Carbon-14 is used to find the age of archaeological objects.

(c) Any two points (1 mark each):

Solution: a homogeneous mixture; particles are very small and do not settle; transparent; passes through filter paper (components cannot be separated by filtration). E.g. salt in water.

Suspension: a heterogeneous mixture of a liquid and a solid; solid particles are larger than 10^{-4} m and settle down; light cannot pass through it; solid is left on filter paper and can be separated by filtration. E.g. saw dust in water.

(d) N H → valencies 3 1 → cross multiply: N_1H_3 → the formula is NH_3 (ammonia). (Method 1 mark; formula 1 mark)

Q.6 Answer in detail (any 1) (5 marks) **[5]**

(a) Bohr's model. Introduction (1 mark): Rutherford's model could not explain the stability of the atom. Niels Bohr put forth the stable orbit atomic model in 1913.

Postulates (3 marks, 1 each): (i) Electrons revolve around the nucleus in concentric circular orbits (shells) at certain distances from it. (ii) The energy of an electron is constant while it is in a particular orbit. (iii) When an electron jumps from an inner to an outer orbit it absorbs energy equal to the difference of the energy levels; when it jumps from an outer to an inner orbit it emits the same amount of energy.

Diagram (1 mark): a positively charged nucleus at the centre with concentric circles labelled K, L, M, N shells and electrons marked on them.

(b) Element, compound, mixture. Definitions with examples (3 marks, 1 each): Element – matter whose smallest particles are all alike atoms of one kind, e.g. oxygen / iron. Compound – matter whose molecules are formed by chemical bonding of two or more types of atoms in a fixed proportion, e.g. water / carbon dioxide. Mixture – matter having two or more elements and/or compounds that are not joined by chemical bonds, e.g. air / brass.

Differences, any two (2 marks): Compound has constant proportion of constituents, mixture has variable proportion; properties of a compound differ from its constituents, in a mixture the constituents keep their properties; constituents of a compound are joined by chemical bonds, those of a mixture are not; all molecules of a compound are alike, a mixture has two or more types of molecules.